



DEEP LEARNING INDABA SOUTH SUDAN 2026

EMPOWERING SOUTH SUDAN'S AI FUTURE



INDABAX SOUTH SUDAN 2026

Inside AI

How it works, what it can do, and where it falls short

Poni Henry · Thursday, 1 October 2026 · Scenius Hub, Juba

TODAY

Our journey in 45 - 60 minutes

- 1 What is AI?
- 2 How machines learn
- 3 Neural networks and deep learning
- 4 How ChatGPT and LLMs work
- 5 Live demo
- 6 Ethics of AI and its shortfalls
- 7 Q&A and where to start

PART 1

What is AI?

A simple definition, and the AI you already use every day

DEFINITION

What is Artificial Intelligence?

AI is the science of making computers do tasks that normally need human intelligence: seeing, hearing, understanding language, learning and making decisions.

Narrow AI (what we have today)

Very good at one task: recognising faces, translating text, recommending videos or answering questions. It cannot step outside what it was built for.

General AI (not here yet)

A machine that could learn and reason across any task like a person. Researchers debate if and when it will arrive. Today it remains an idea, not a product.

QUICK QUESTION

Who has used AI today?

Google Maps traffic

Face unlock

WhatsApp auto-correct

Spam filters

Social media feeds

ChatGPT

If you used any of these, **you used AI.**

CAPABILITIES

What can AI do?

See

Recognise faces, read number plates, spot disease in crop leaves and X-rays.

Hear

Turn speech into text and understand voice commands.

Understand & write

Translate languages, summarise documents and answer questions.

Predict

Forecast rain, crop yields, traffic or which customers may stop paying.

Recommend

Choose the next video, song or product you are likely to want.

Create

Generate text, images, music and video from a simple prompt.

How can South Sudan use AI?

AI can...	How South Sudan could use it	Sector
See	Spot crop and cattle diseases from a phone photo; track floods from satellite images, like Bentiu Flood Watch	Farming, disasters
Hear	Voice assistants in Juba Arabic, Dinka or Nuer for people who cannot read or type	Inclusion, services
Understand & write	Translate health, legal and government information into local languages	Health, justice
Predict	Forecast floods, dry spells, crop yields and disease outbreaks such as cholera	Food security, health
Recommend	Personalised lessons for students; match young people with jobs and training	Education, jobs
Create	Produce learning materials and radio content in local languages, fast and cheaply	Education, media

A SHORT HISTORY

70 years of AI

1950

Alan Turing asks: can machines think?

1956

The term “Artificial Intelligence” is coined at Dartmouth

1997

IBM's Deep Blue beats chess champion Garry Kasparov

2012

Deep learning makes a leap in recognising images

2016

AlphaGo beats a world champion at the game of Go

2022

ChatGPT launches and brings AI to millions

AI is not new. What changed recently is more data, faster computers and better algorithms.

CLOSE TO HOME

AI in Africa and South Sudan

African languages

Masakhane, a grassroots research community, builds AI tools for African languages, so AI understands our voices too.

Farming

Phone apps help farmers photograph crop leaves and spot diseases early, before a whole harvest is lost.

Finance

Banks and mobile money services use AI to detect fraud and assess credit for people with no formal records.

South Sudan

Local innovators are building too: flood early warning for Bentiu, school systems like AcademiaPro, and today's hackathon projects.

The question is not whether AI will reach South Sudan. It is who will build it.

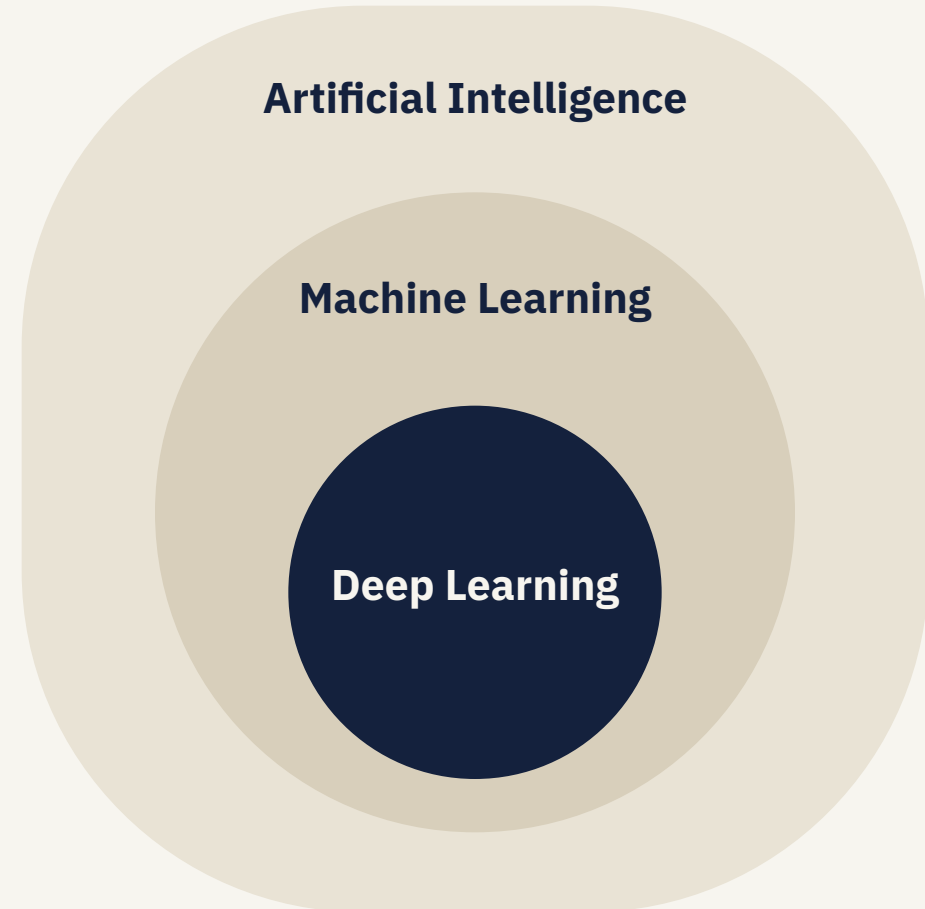
DEFINITIONS

AI, ML and deep learning

AI: computers doing tasks that normally need human intelligence.

Machine learning: the part of AI where computers learn from data instead of being given rules.

Deep learning: machine learning with large neural networks.



THE BIG SHIFT

Rules vs learning

Traditional programming

A person writes the rules, and the computer applies them to data.

Rules + data → answers

Machine learning

We give the computer examples with answers, and it works out the rules itself.

Data + answers → rules

EXAMPLES

Rules or learning?

Task	Rules way (a person writes the rule)	Machine learning way (learns from examples)
Blocking spam	Block any email containing the word “lottery”	Learns from millions of emails people marked as spam
Unlocking your phone	Type the PIN you set, such as 1234	Face unlock learns what your face looks like
Translating text	Swap each word using a dictionary	Google Translate learns from millions of translated sentences
Approving a loan	Approve if salary is above a fixed amount	Learns from past customers who did and did not repay
Adding up a bill	Calculator: add the numbers. Rules work perfectly	Not needed. Simple, fixed tasks stay with rules

PART 2

How machines learn

Data, training, testing, and three main types of learning

What is Machine Learning?

Machine learning (ML) is about designing algorithms that automatically extract useful information from data ([mathematics for machine learning](#)). Instead of explicitly programming every rule, ML allows the computer to learn from data and make predictions or decisions based on what it has learned.

In ML, there are three main components: data, models, and learning.

Key Components

Data

These are generally a collection of examples that we want to learn from.

Models

Functions that take data as input and produce predictions or decisions as output. These models are controlled by parameters, which are like adjustable knobs that determine how the model behaves.

Learning

The process of adjusting the model's parameters to improve its performance on a specific task.

1.1 Data

ML is about finding patterns in data, which can take various forms such as images, text, audio, and more. For computers to process and analyze this data, it must be represented numerically, often as vectors (which are like lists of numbers) or matrices (which are like grids of numbers). For example, an image can be represented as a grid where each cell contains a pixel value, reflecting the brightness or colour at that point, the number 0 can be used for black and 255 for white, and the numbers in between can represent varying brightness from black to white.

Understanding Inputs and Labels:

In a machine learning task, our data typically consists of two main components: inputs (also called features) and labels. The input is the information we have and use to make predictions. The label is the outcome we want to predict.

For example, in a digit recognition task:

- The **input** would be an image of a handwritten digit.
- The **label** would be the actual digit that the image represents (e.g., 0, 1, 2, ..., 9).

 **Task: Open Colab for a practical example.**

[Copy of Introduction to Deep Learning.ipynb](#)

1.2 Models

Models transform data in useful ways. They take input data and produce the desired output.

Formally, a model is a function $f(x)$ that maps input data x (a sample from our dataset) to output data y (the label). For example, in our crop yield dataset, the input is a vector of different measurements from the farm such as rainfall, temperature and fertiliser use, and the output is the predicted crop yield.

The model's behaviour is controlled by **parameters**, which are adjustable values updated during the learning process.

Popular machine learning models

Model	Type	How it works, simply	Example use
Linear regression	Supervised: numbers	Draws the best straight line through the data	Predicting crop yield from rainfall
Logistic regression	Supervised: yes / no	Estimates the chance of an outcome	Will a customer repay a loan?
Decision tree	Supervised	Asks a chain of yes/no questions	Deciding if a patient needs referral
Random forest	Supervised	Many trees vote; the majority wins	Our live demo: crop yield
XGBoost	Supervised	Builds trees one by one, each fixing the last one's mistakes	Top choice in Zindi and Kaggle contests
K-nearest neighbours	Supervised	Looks at the most similar past examples	Recommending similar products
K-means	Unsupervised	Groups data points that sit close together	Grouping customers by buying habits
Neural network	Supervised and more	Layers of connected units learn complex patterns	Face unlock, voice typing, ChatGPT

1.3 Learning

Learning is the process of improving a model's performance by updating its parameters. This improvement is guided by an **objective function**, often called the **loss function**, which measures how well the model is performing.

Loss Function

The **loss function**, denoted as L , quantifies the difference between the model's prediction $f(x)$ and the actual outcome y . It outputs a number that represents how close the prediction is to the actual outcome.

- **Prediction:** $f(x)$
- **Actual Outcome:** y
- **Loss Function:** $L(f(x), y)$

The goal of learning is to **minimize** this loss function. Although we could also maximize the loss function in theory, minimizing it is the conventional approach.

Learning Algorithm

The **learning algorithm** is responsible for adjusting the model's parameters to minimize the loss function. Intuitively, you can think of learning as **tuning** the model's knobs (parameters) to get the desired behaviour. The learning algorithm does the tuning, while the loss function tells us *how well the tuning is working*.

Formally, learning involves finding the parameters θ of our model f that minimize the loss function L over the dataset X :

Minimize $L(f(x; \theta), y)$ over X .

Generalization

Our goal is not only to perform well on the training data but also to ensure that the model generalizes well to new, unseen data — for example, a farm or a season the model has never seen.

Learning can occur in various settings. We will explore the main types of learning in the next section.

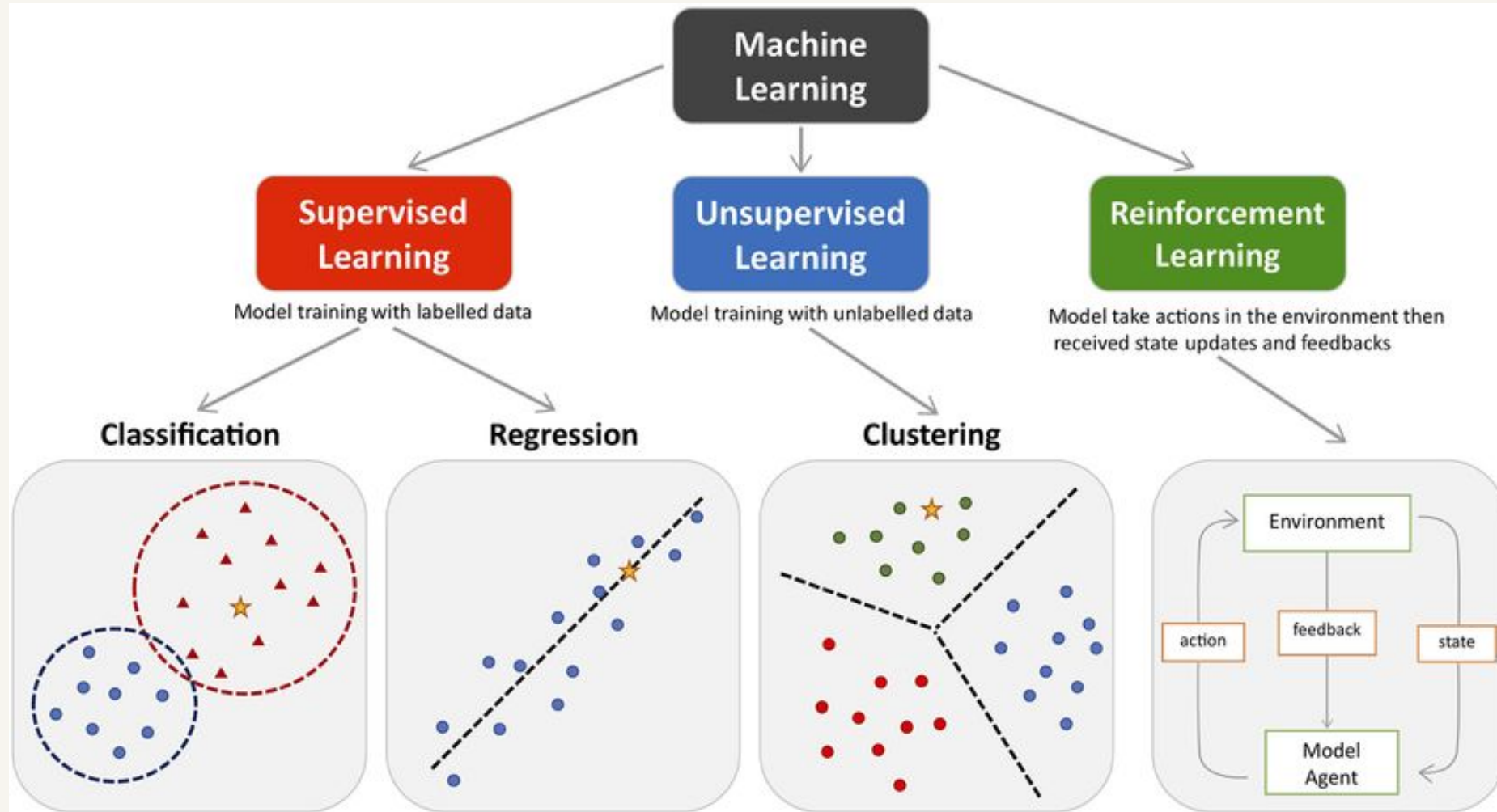
TRAINING

How a model learns from mistakes



We keep some data hidden for **testing**, like secret exam questions. A good model works on data it has never seen, not just the examples it memorised.

Types of Machine Learning Problems



1.4 Types of machine learning problems

Broadly speaking, machine learning tasks can be classified into three categories.

Supervised Learning

In supervised learning, the algorithm is trained on a labelled dataset, where each input data point is paired with the correct output (label). The goal is to learn a mapping from inputs to outputs based on the training data. Once trained, the model can make accurate predictions on new, unseen data. This approach is commonly used for tasks like classification and regression, where the correct answers are known during training.

Unsupervised Learning

In unsupervised learning, the algorithm works with an unlabeled dataset, meaning the data has no predefined labels or outcomes. The objective is to discover patterns, structures, or relationships within the data. This might involve grouping similar data points (clustering) or reducing the dimensionality of the data to uncover hidden patterns. Unsupervised learning is often used in exploratory data analysis and for finding natural groupings in data.

Reinforcement Learning

*In reinforcement learning, the algorithm learns to make decisions by interacting with an environment. The learner, known as the *agent*, takes actions and receives feedback in the form of rewards or penalties. The goal is to learn a strategy, or *policy*, that maximizes the cumulative reward over time. Reinforcement learning is commonly applied in areas like robotics, game playing, and autonomous systems, where an agent must learn to navigate complex environments.*

EVERYDAY EXAMPLES

Where each type is used

Supervised

Learns from examples with answers

- Face unlock and photo tagging
- Spam filters
- Voice typing: speech to text
- Spotting crop disease from leaf photos
- Predicting crop yield or exam results

Unsupervised

Finds groups and patterns with no answers given

- Grouping customers by buying habits
- Google Photos grouping pictures of the same person
- News apps grouping similar stories
- Flagging unusual mobile money transactions
- Spotting new trends in social media posts

Reinforcement

Learns by trial, reward and penalty

- Game AI that beat world champions at chess and Go
- Robots learning to walk and pick things up
- Self-driving cars learning to drive safely
- ChatGPT improved with human feedback on its answers
- Delivery apps learning the fastest routes

LEARNING APPROACHES

How models can learn

Supervised

Learns from labelled examples: inputs with the right answers.

Unsupervised

Finds groups and patterns in data with no answers given.

Reinforcement

Learns by trial and error, guided by rewards and penalties.

Semi-supervised

Uses a few labelled examples plus many unlabelled ones. Useful when labelling is costly.

Self-supervised

Creates its own labels from raw data, such as hiding a word and predicting it. This is how ChatGPT-style models are pre-trained.

Transfer learning

Starts from a model already trained on a big dataset, then fine-tunes it for a new task with little data.

Transfer learning matters for South Sudan: we can adapt big global models to local languages and data without starting from zero.

PART 5

Live demo

Build and train a model in a few lines of Python

IN GOOGLE COLAB

Predict a farm's crop yield

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor

X = data[["rainfall_mm", "temperature_c", "fertiliser_kg"]]
y = data["yield_tonnes"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = RandomForestRegressor()
model.fit(X_train, y_train) # learning happens here
```

1. Load data
2. Split train / test
3. Train with fit
4. Test and predict

https://colab.research.google.com/drive/1XMYuu7cs6XB_zECGxzSbY4Movc78N_MM?usp=sharing

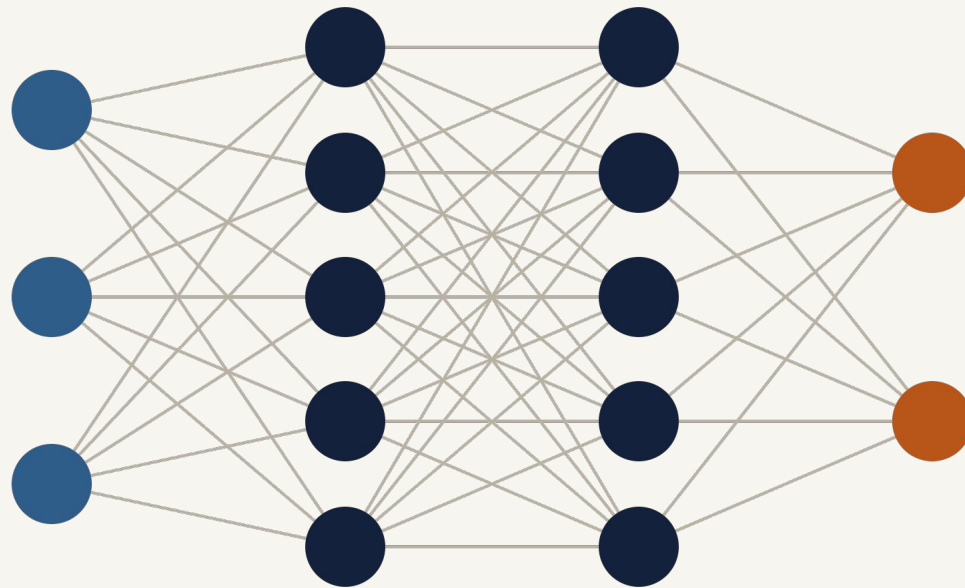
PART 3

Neural networks and deep learning

Simple units, connected in layers, learning complex patterns

NEURAL NETWORKS

Many simple units, connected in layers



Input

Hidden layers

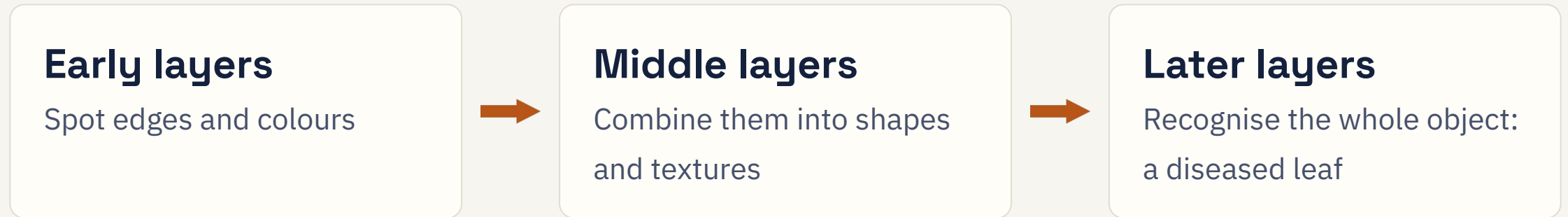
Output

Neuron: takes numbers in, weighs how important each one is, adds them up and passes the result on.

Layers: input (the data), hidden (where learning happens), output (the answer).

Learning: after each mistake, every weight is nudged slightly. This is called backpropagation.

Why deep networks can see



Example: a farmer photographs a cassava or sorghum leaf, and the app identifies the disease. Similar models identify snake species from photos to guide treatment.

Types of neural networks

Feedforward (MLP)

The basic network: data flows straight from input to output through layers.

e.g. predicting crop yield or loan risk from a table

CNN (Convolutional)

Scans images in small patches to find edges, shapes and objects.

e.g. face unlock, spotting crop disease in leaf photos

RNN / LSTM

Reads data in order and remembers what came before.

e.g. early speech recognition, forecasting rainfall over time

Transformer

Uses “attention” to weigh every word against every other word at once.

e.g. ChatGPT, Google Translate. The T in GPT

GAN (Generative

Adversarial)
Two networks compete: one creates fakes, the other spots them.

e.g. realistic faces and deepfakes

Diffusion model

Learns to turn random noise step by step into a clear image.

e.g. AI image generators like Midjourney

PART 4

How ChatGPT and LLMs work

Predicting the next word, at enormous scale

THE CORE TRICK

Predict the next word

The capital of South Sudan is **Juba**

The model writes whole answers one word (token) at a time.

Pre-training

Learns from billions of pages of text: grammar, facts and patterns of reasoning.

Fine-tuning

Further trained with human feedback to be helpful and follow instructions.

What LLMs can and cannot do

Can

Explain, summarise and translate

Draft emails, essays and code

Brainstorm ideas quickly

Cannot (reliably)

Guarantee facts: they can hallucinate

Truly understand what they say

Work well in most South Sudanese languages

PART 6

Ethics of AI and its shortfalls

Powerful, but imperfect: using AI responsibly

SHORTFALLS

Where AI gets it wrong

Shortfall	What it means	Example
Bias	Repeats unfair patterns in its data	A loan model trained on urban customers rejects rural applicants
Hallucinations	States false information confidently	ChatGPT inventing a source or statistic
Poor local fit	Models trained elsewhere fail here	A crop app misreading South Sudanese varieties
Language exclusion	Little data in local languages	Voice tools that fail outside English and Arabic
No real understanding	Finds patterns but lacks common sense	Fooled by small image changes a human would ignore

ETHICS

Risks we must manage

Privacy and consent

Who collected the data, and did people agree?

Deepfakes

Fake audio and video spread fast, especially in elections

Accountability

When AI causes harm, who is responsible?

Jobs and skills

Some tasks will be automated; build skills to work with AI

Honesty

Use AI to learn, not to pass off its work as yours

Data sovereignty

African data should benefit African communities

The gaps in AI and ML in South Sudan

Local data

Very little digitised data about South Sudan, and almost none in local languages like Dinka, Nuer, Bari or Juba Arabic.

Connectivity and power

Internet is expensive and patchy, and power cuts make it hard to learn, build and run AI tools.

Computing power

Few GPUs or data centres. Cloud tools often need international cards that many people cannot get.

Skills and training

Few university courses or trainers in AI and data science, and limited hands-on practice.

Policy and protection

No national AI strategy yet, and limited rules to protect people's data and guard against misuse.

Funding and ecosystem

Little investment in local AI startups and research, and few jobs that use these skills yet.

Every gap is also an opportunity for the people in this room to build, research and lead.

WHAT WE CAN DO

Responsible AI is everyone's job

Builders and organisations

- 1 Use local, representative data
- 2 Always check AI outputs
- 3 Protect people's data and ask for consent
- 4 Keep humans in the loop for big decisions
- 5 Build AI for South Sudanese languages and communities

Students

- 1 Use AI to learn, not to copy: do your own thinking
- 2 Check facts before you share or submit them
- 3 Never put personal data or others' photos into AI tools
- 4 Don't create or spread deepfakes and fake news
- 5 Build projects that solve problems in your community

Discuss: name one place in South Sudan where AI could help, and one way it could cause harm.

WHERE TO START

Free resources to keep learning

MIT Learn

Free MIT courses on AI and ML

learn.mit.edu

Google Colab

Free Python notebooks with GPU

colab.research.google.com

Kaggle Learn

Short, hands-on ML courses

kaggle.com/learn

Zindi

African data science competitions

zindi.africa

Masakhane

NLP for African languages

masakhane.io

Scenius Hub

Local meetups and hackathons in Juba

[Juba](#)

OUR COMMUNITY

The Deep Learning Indaba

A movement to strengthen African machine learning and AI, run by Africans, for Africa.

- *Indaba* means a gathering to discuss important matters in isiZulu
- Started in 2017 with its first annual meeting in Johannesburg
- Held in a different African country each year, including Kenya, Tunisia, Ghana, Senegal and Rwanda
- Brings together students, researchers and industry to learn, share and build

IndabaX

Locally organised events in countries across Africa.
Today's event is IndabaX South Sudan 2026.

Awards and mentorship

Programmes like the Kambule and Maathai awards celebrate African research and impact.

How to join

Follow deeplearningindaba.com and apply to attend the next Indaba.

PRACTISE AND COMPETE

Zindi

Africa's largest data science competition platform, connecting African data scientists with real problems to solve.

- Founded in 2018 and based in Cape Town, South Africa
- Tens of thousands of data scientists across the continent
- Competitions set by companies, NGOs and governments, often with cash prizes
- A place to build a portfolio that employers can see

Compete

Solve real challenges in farming, health, finance and more. Today's hackathon ran on Zindi.

Learn

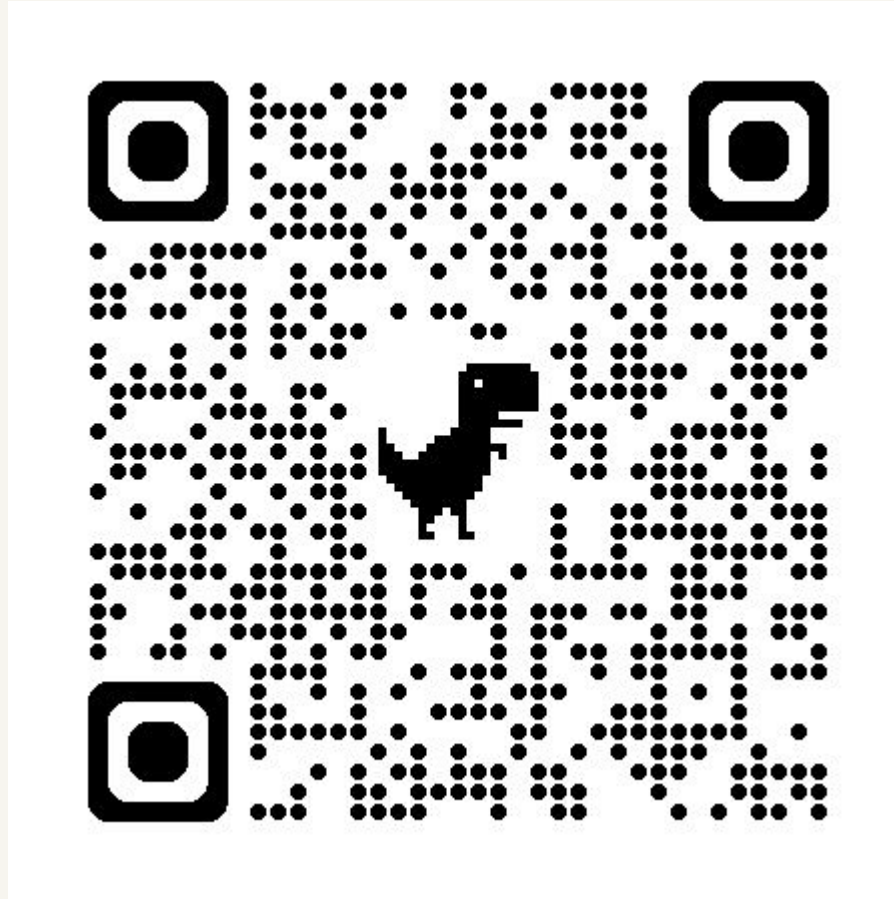
Beginner-friendly challenges and tutorials to practise your first models.

Get started

Sign up free at zindi.africa and join a challenge this week.

PRACTISE AND COMPETE

Learning Resources on Deep Learning indaba



AI could shape South Sudan's future.

Let's make sure South Sudanese help build it. Start today.

Questions?